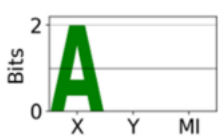
A

SI motif,
Case A

XY

0.25 AA
0.25 AC
0.25 AG
0.25 AT



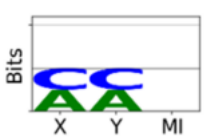
$SI(X) + SI(Y) + MI(X;Y)$
$= \mathbf{2 + 0 + 0}$

B

SI motif,
Case B

XY

0.25 AA
0.25 AC
0.25 CA
0.25 CC



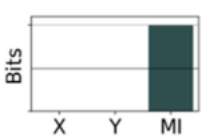
$SI(X) + SI(Y) + MI(X;Y)$
$= \mathbf{1 + 1 + 0}$

C

MI motif

XY

0.25 AC
0.25 CG
0.25 GT
0.25 TA



$SI(X) + SI(Y) + MI(X;Y)$
$= \mathbf{0 + 0 + 1}$

$$TI = log_2\left(\frac{n_{total}}{n_{selected}}\right) = log_2\left(\frac{16}{4}\right) = \mathbf{2} \quad (bits)$$

D

